

The Pair, the Triad and the Half: One Pattern in Physical, Biological and Learning Systems

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Abstract

A pair is two states exchanged by a mirror, and its point of balance is the half. A triad is three states exchanged by a turn, and its generating function $z^{-1} + 1 + z$ vanishes at one third and two thirds of a turn. We state this pattern exactly in eight systems and compute each instance at its own size. A balanced three-phase current lies in the pair of turning sequences and needs no return wire. The coherence of a carbon nucleus coupled to a deuteron is the character of the triad and vanishes at one third and two thirds of the coupling period. Two spins one half split into a triad and a fourth state. Around a planet the two collinear Lagrange points differ by a third of the Hill scale: we derive the exact series $\mathcal{A} = \frac{h}{3}(1 - \frac{h^2}{27} - \frac{h^3}{3} + O(h^4))$ for their asymmetry and evaluate it for the giant planets and for the Earth with the Moon. The sodium conductance of the squid axon is three activation gates and one inactivation gate, and its gate is balanced at -40.02 mV, where its entropy is greatest. At the self-dual radius a closed string acquires the triad of charges $-1, 0, 1$. The cloud tops of the giant planets fall on the two sides of the titration zero $N/S = 1$. In ternary quantisation of weights the least-squares threshold is one third of the range for uniform weights, setting exactly one third of them to zero.

Keywords: triad; symmetrical components; nuclear magnetic resonance; spin; Lagrange points; Hill sphere; Hodgkin–Huxley model; T-duality; giant planet atmospheres; ternary neural networks

1. Introduction

Some patterns are older than the fields in which they are found. Two states exchanged by a mirror have a point of balance halfway between them. Three states exchanged by a turn have a sum that survives the turn and two combinations that the turn rotates. The second statement is exact in a single line of algebra: the generating function $z^{-1} + 1 + z$ of the three states $-1, 0, 1$ vanishes at $z = e^{\pm 2\pi i/3}$, at one third and two thirds of a turn, and nowhere else.

This paper states the pattern in eight systems and computes each instance. The systems come from electrical engineering, nuclear magnetic resonance, quantum mechanics, celestial mechanics, the physiology of nerve, string theory, planetary chemistry and machine learning. For each we give the exact statement and the number that it fixes. We claim no link between the systems beyond the shared algebra. Where an instance is a classical fact, we say so and credit it. The new result is the third-order law for the Lagrange points (Theorem 1).

2. The Pattern

A *pair* is a space with an involution σ , $\sigma^2 = 1$. It splits into the even part ($\sigma = 1$) and the odd part ($\sigma = -1$), and a two-state system with occupation p is balanced at $p = \frac{1}{2}$. There its binary entropy $-p \log p - (1-p) \log(1-p)$ takes its maximum $\log 2$, and its variance $p(1-p)$ its maximum $\frac{1}{4}$.

A *triad* is a space with a turn C , $C^3 = 1$. On three states it is the cyclic shift, and

$$J = I + C + C^2$$

is the matrix of ones. J multiplies the *still pattern* $(1, 1, 1)$ by 3 and annihilates the two *turning patterns* $(1, \omega, \omega^2)$ and $(1, \omega^2, \omega)$, $\omega = e^{2\pi i/3}$. On the unit circle the generating function of the triad is

$$z^{-1} + 1 + z = 1 + 2 \cos \theta = \chi_1(\theta),$$

the character of the spin-one representation of $SU(2)$ at rotation angle θ [13]. It vanishes exactly at $\theta = \pm \frac{2\pi}{3}$ (Figure 1a). A *fourth* is a further state that reads the triad's sum, so that three and one together make four.

3. Polyphase Currents

Fortescue decomposed a three-phase system into a zero sequence and a positive and a negative sequence [5]. In the terms of Section 2, the projection onto the zero sequence is $\frac{1}{3}J$, and the positive and negative sequences are the two turning patterns. A balanced three-phase system, such as the currents of Tesla's polyphase motor [15], lies entirely in the turning patterns. Its zero sequence, the sum of the three currents, vanishes because $1 + \omega + \omega^2 = 0$, which is why a balanced three-phase load needs no return wire.

4. A Nucleus Coupled to a Deuteron

A spin- $\frac{1}{2}$ nucleus such as ^{13}C , scalar-coupled with constant $\mathcal{J}$ to a deuteron, a spin-one nucleus whose three states $m = -1, 0, 1$ are equally populated, precesses at three frequencies spaced by $\mathcal{J}$: the 1:1:1 triplet [10]. Its coherence is modulated by

$$\frac{1}{3}(1 + 2 \cos 2\pi \mathcal{J}t) = \frac{1}{3}\chi_1(2\pi \mathcal{J}t),$$

which vanishes exactly at $t = 1/(3\mathcal{J})$ and $t = 2/(3\mathcal{J})$ in every period (Figure 1b). The silences of the coherence are the zeros of the triad.

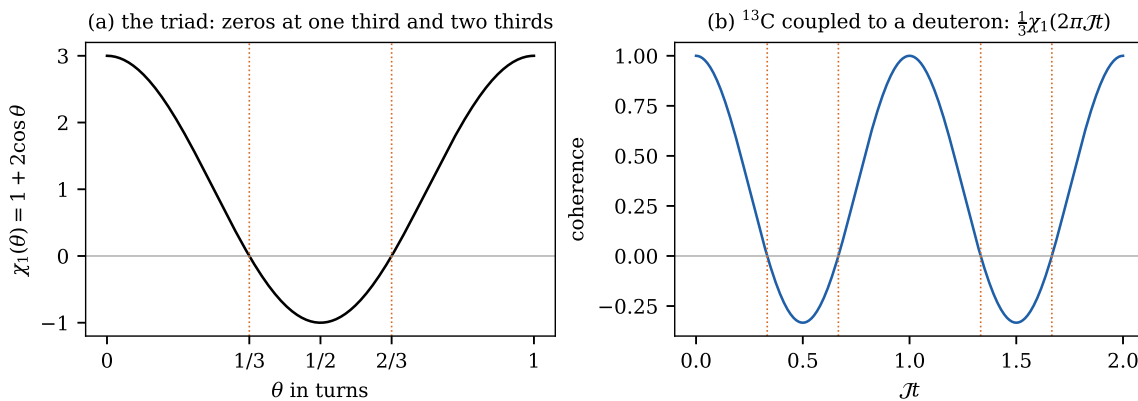


Figure 1. (a) The character $\chi_1(\theta) = 1 + 2 \cos \theta$ of the triad, zero at one third and two thirds of a turn. (b) The coherence of a ^{13}C nucleus coupled to a deuteron, $\frac{1}{3}\chi_1(2\pi \mathcal{J}t)$, silent at $\mathcal{J}t = \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \frac{5}{3}$.

5. Two Spins: A Triad and a Fourth

On two spins $\frac{1}{2}$ the exchange of the spins is an involution of $\mathbb{C}^2 \otimes \mathbb{C}^2$. Its eigenvalue $+1$ belongs to the three-dimensional triplet, with $m = -1, 0, 1$, and its eigenvalue -1 to the one-dimensional singlet [13]:

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0.$$

The pair splits into a triad and a fourth. The triad is the spin-one representation whose character is χ_1 , and the fourth is the state that the exchange reverses.

6. The Two Lagrange Throats

In the circular restricted three-body problem with mass ratio μ , let γ_1 and γ_2 be the distances of the collinear points L_1 and L_2 from the smaller body, in units of the separation. They are the positive roots of the classical quintics [14]

$$\gamma^5 \mp (3 - \mu)\gamma^4 + (3 - 2\mu)\gamma^3 - \mu\gamma^2 \pm 2\mu\gamma - \mu = 0,$$

with the upper signs for L_1 and the lower for L_2 . Let $h = (\mu/3)^{1/3}$ be the Hill scale, and let

$$\mathcal{A} = \frac{\gamma_2 - \gamma_1}{\gamma_2 + \gamma_1}$$

be the asymmetry of the two throats.

Theorem 1 (the throats differ by a third). As $\mu \rightarrow 0$,

$$\gamma_{1,2} = h \mp \frac{1}{3}h^2 - \frac{1}{9}h^3 + O(h^4), \quad \mathcal{A} = \frac{h}{3} \left(1 - \frac{h^2}{27} - \frac{h^3}{3} + O(h^4) \right).$$

Proof. Put $\mu = 3h^3$ and $\gamma = h(1 + a_1h + a_2h^2 + \dots)$ into each quintic and solve for the coefficients order by order. The coefficient of h^4 fixes a_1 , and each further order is linear in the next coefficient. This gives

$$\gamma_1 = h - \frac{1}{3}h^2 - \frac{1}{9}h^3 + \frac{58}{81}h^4 + O(h^5), \quad \gamma_2 = h + \frac{1}{3}h^2 - \frac{1}{9}h^3 + \frac{50}{81}h^4 + O(h^5).$$

Then $\gamma_2 - \gamma_1 = \frac{2}{3}h^2 - \frac{8}{81}h^4 + O(h^5)$ and $\gamma_2 + \gamma_1 = 2h - \frac{2}{9}h^3 + \frac{4}{3}h^4 + O(h^5)$, and the quotient expands to the stated series. The computation was done exactly in SymPy, and the two correction coefficients were confirmed numerically from the roots of the quintics: -0.037039 at $\mu = 10^{-15}$ for the second, and -0.3333333818 at $\mu = 10^{-18}$ for the third. $\square$

The leading term is the classical statement that the two throats lie at nearly equal distances, on the scale h , around the smaller body. The theorem adds that their difference is $\frac{1}{3}$ of that scale, corrected first by $-\frac{1}{27} = -(\frac{1}{3})^3$ and then by $-\frac{1}{3}$ at third order. The third-order term equals $-\frac{h^3}{3} = -\frac{\mu}{9}$. With the mass parameters of the ephemeris DE440 [4], Table 1 gives the exact asymmetry, from the roots of the quintics, against the series.

Table 1. The asymmetry of the throats L_1 and L_2 of the giant planets, and of the Earth with the Moon, about the Sun.

	μ	h	$\mathcal{A}$	$\mathcal{A}/(h/3)$	$\mathcal{A}/(\frac{h}{3}(1 - \frac{h^2}{27}))$
Jupiter	$9.5388 \cdot 10^{-4}$	0.068253	0.022745	0.99972	0.99989
Saturn	$2.8580 \cdot 10^{-4}$	0.045672	0.015222	0.99989	0.99997
Uranus	$4.3661 \cdot 10^{-5}$	0.024415	0.0081381	0.99997	0.999995
Neptune	$5.1511 \cdot 10^{-5}$	0.025798	0.0085992	0.99997	0.999994
Earth+Moon	$3.0404 \cdot 10^{-6}$	0.010045	0.0033482	0.999996	0.9999997

In the last column the remaining factor is $1 - \frac{\mu}{9}$ to the next order: 0.999894 for Jupiter, against 0.999893 computed.

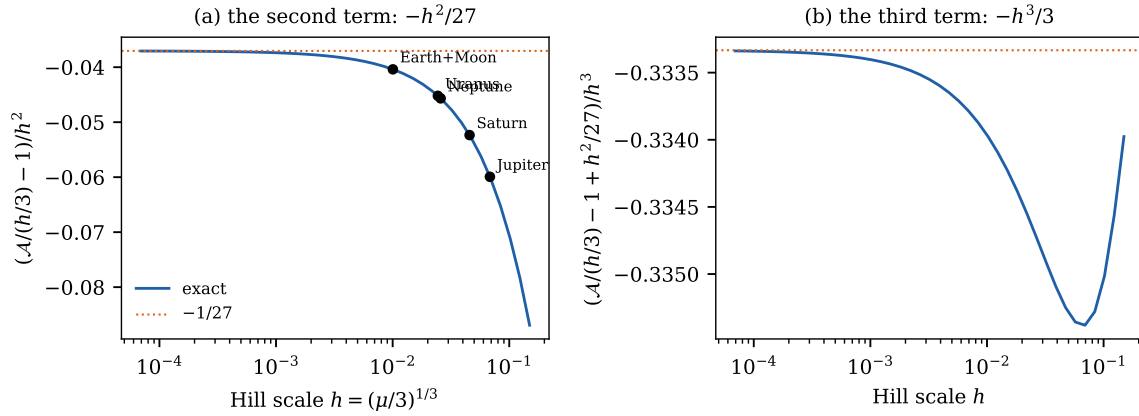


Figure 2. (a) The second-order coefficient of the asymmetry, from the roots of the quintics, tending to $-\frac{1}{27}$ as $h \rightarrow 0$, with the planets marked. (b) The third-order coefficient, tending to $-\frac{1}{3}$.

7. The Sodium Gate

Hodgkin and Huxley wrote the sodium conductance of the squid giant axon as $\bar{g}_{\text{Na}} m^3 h$: three activation gates m and one inactivation gate h , each a two-state system with its own voltage-dependent rates [7]. The channel is a triad and a fourth. The channel protein has four homologous domains [1]. Optical tracking of their voltage sensors shows three of them moving with activation and the fourth, more slowly, with inactivation [2].

Each activation gate is a pair with open probability $m_{\infty}(V) = \alpha_m / (\alpha_m + \beta_m)$. With the rate functions of [7] in the modern convention, with the resting potential at -65 mV [3], namely $\alpha_m = 0.1(V + 40) / (1 - e^{-(V+40)/10})$ and $\beta_m = 4e^{-(V+65)/18}$ in ms^{-1} , the gate is balanced, $m_{\infty} = \frac{1}{2}$, at

$$V = -40.02 \text{ mV}.$$

At that voltage the entropy of the gate takes its maximum $\log 2$ and its variance its maximum $\frac{1}{4}$ (Figure 3).

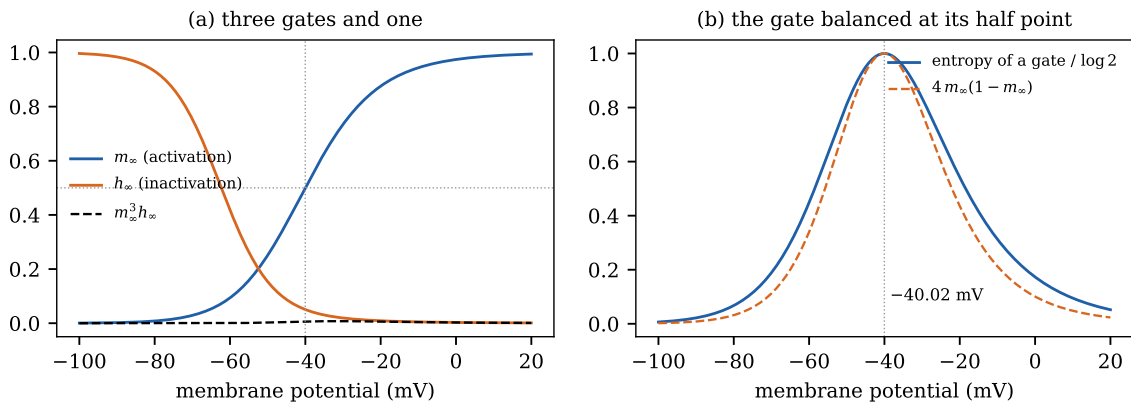


Figure 3. (a) The steady states of the activation gate m_{∞} , the inactivation gate h_{∞} , and the open fraction $m_{\infty}^3 h_{\infty}$. (b) The entropy and the variance of an activation gate, both greatest where the gate is balanced, at -40.02 mV.

8. The Self-Dual Radius

A closed string on a circle of radius R has momentum n and winding w , and $p_{L,R} = n/R \pm wR/\alpha'$. T-duality $R \mapsto \alpha'/R$ exchanges n and w and reverses p_R : it is a pair, with its fixed point at $R = \sqrt{\alpha'}$, the midpoint of the duality on the logarithmic scale of R . There the states with $n = w = \pm 1$ and $p_R = 0$ become massless gauge bosons. They enhance $U(1) \times U(1)$ to $SU(2) \times SU(2)$, and the charges of each $SU(2)$ are $-1, 0, 1$, the triad [12, §8.3], [6].

9. The Titration Zero of the Giant Planets

Ammonia and hydrogen sulphide combine one to one into the ammonium hydrosulphide cloud [17], so above that cloud only the species in excess survives. The ratio $N/S = 1$ is the zero of the pair. On Jupiter and Saturn nitrogen is in excess, and the upper clouds are ammonia ice. Above the clouds of Uranus and Neptune hydrogen sulphide has been detected [8,9], so there sulphur is in excess. The four giant planets fall on the two sides of the zero.

10. Ternary Weights

Quantise a weight w to three levels, $w \mapsto \alpha \operatorname{sgn}(w)[|w| > \Delta]$, with α and Δ chosen to minimise the mean squared error.

Proposition 1 (the least-squares threshold). *For a symmetric distribution of w the optimal threshold solves $\Delta = \frac{1}{2} \mathbb{E}[|w| : |w| > \Delta]$. For weights uniform on $[-1, 1]$ it is $\Delta = \frac{1}{3}$, which sets exactly one third of the weights to zero. For Gaussian weights it is $\Delta = 0.6120\sigma$, which sets 45.95% of them to zero.*

Proof. For fixed Δ the best α is $\mathbb{E}[|w| : |w| > \Delta]$, and the error is then $\mathbb{E}[w^2] - S(\Delta)^2/P(\Delta)$ with $S = \mathbb{E}[|w|; |w| > \Delta]$ and $P = \Pr(|w| > \Delta)$. Since $S' = -2\Delta\rho(\Delta)$ and $P' = -2\rho(\Delta)$ for the density ρ , the derivative vanishes where $\Delta = S/(2P) = \alpha/2$. For the uniform law this reads $\Delta = \frac{1+\Delta}{4}$, so $\Delta = \frac{1}{3}$ and $\Pr(|w| \leq \Delta) = \frac{1}{3}$. The Gaussian case is solved numerically. $\square$

The threshold near 0.6σ is the rule of ternary weight networks [16]. For uniform weights the zero level, the still pattern of the three levels, takes exactly a third (Figure 4).

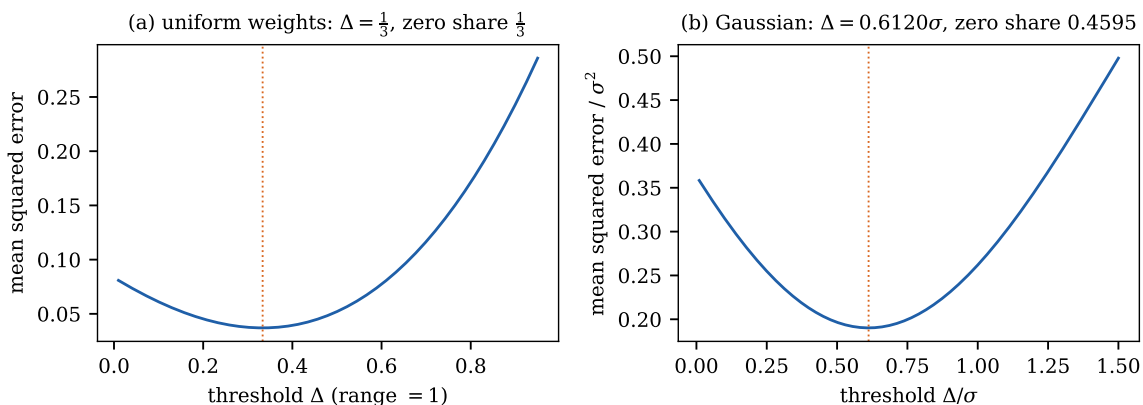


Figure 4. The mean squared error of ternary quantisation against the threshold. (a) Uniform weights: minimum at $\Delta = \frac{1}{3}$, one third set to zero. (b) Gaussian weights: minimum at $\Delta = 0.6120\sigma$, 45.95% set to zero.

11. Methods

The Lagrange series in SymPy; the roots of the quintics with mpmath [11] at 40 to 60 digits; the mass parameters GM of DE440 for the Sun ($132712440041.279419 \text{ km}^3\text{s}^{-2}$) and for the planetary systems. The half point of the sodium gate by bracketing the root of $\alpha_m = \beta_m$. The ternary thresholds by bracketing the root of $\Delta - \frac{1}{2}\mathbb{E}[|w| : |w| > \Delta]$. The scripts are provided with this article.

12. Conclusions

A pair balances at its half, and a triad keeps its sum and is silent at its thirds. The pattern is exact in each of the eight systems here.

- A balanced three-phase current lies in the turning sequences.
- The coherence of a nucleus coupled to a deuteron falls silent at the thirds of its period.
- Two spins split into a triad and a fourth.
- The Lagrange throats differ by a third of the Hill scale, with the corrections $-\frac{h^2}{27}$ and $-\frac{h^3}{3}$.

- The sodium gate is three and one, and balances at -40.02 mV.
- The self-dual string carries the charges $-1, 0, 1$.
- The giant planets fall on the two sides of the titration zero.
- Uniform weights set a third of themselves to zero.

Each statement is made at the size of its result. Together they show one small piece of algebra at work in very different matter.

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